



Corrosion Inhibition Performance Of An Expired Drug On Corrosion Of Copper In Nitric And Sulphuric Acid Media

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ABSTRACT

Aggregate behaviour Amisulpride, an expired anti-hypertensive drug, was explored to determine its corrosion inhibition performance in 0.5 M H₂SO₄ and 1.0 M HNO₃ solutions on corrosion of copper through weight-loss, electrochemical impedance spectroscopy (EIS), potentiodynamic polarization, and quantum chemical calculations. The experimental results indicated that Amisulpride has a high level of adsorption and corrosion inhibition capacity both acid solutions. The maximum corrosion inhibition efficiencies of 95.3% and 88.3% at 1.0 mM were found in 0.5 M H₂SO₄ and 1.0 M HNO₃ solutions respectively, which is due to the protection layer formed on the copper surface remaining intact. The addition of inhibitors to the corrosive media led to higher values of charge-transfer resistance and lower values of the double-layer capacitance which indicated the formation of a surface films. A study on polarization revealed that Amisulpride is a mixed-type inhibitor; but dominates the cathodic reaction. The adsorption was performed by following the Langmuir isotherm and thermodynamic parameters proved to be spontaneous, highly physisorptive process with a small contribution of chemisorption. Studies on the effect of temperature revealed that the activation energy decreases in the presence of the inhibitor indicating an increased coverage on surfaces at high temperatures. Quantum chemical analyses were used to confirm the experimental observation, which emphasized positive electronic features and active adsorption centres on both neutral and protonated amisulpride. The study proves that the expired Amisulpride drug is viable and environmental enthusiastic compound in sulfuric and nitric acid as an inhibitor of copper corrosion.

Keywords: Corrosion inhibitors; Copper corrosion- Electrochemical impedance spectroscopy (EIS)- Potentiodynamic polarization- Molecular reactivity parameters.

1. INTRODUCTION

Corrosion of copper and its alloys is an important problem in the industrial sector since the material has a broad application in electrical equipment, coolers, chemical work, and heat-exchange equipment (Verma *et al.* 2017). Even though copper has a distinctive capacity to create adherent oxide layers, the layers are not stable in highly oxidizing or acidic conditions and therefore tend to dissolve rapidly (Kaya *et al.* 2016). Sulphuric and nitric acids are also mineral acids that are frequently used in industrial cleaning, pickling, etching and hydrometallurgical processes and so control of copper corrosion in mineral acids is highly significant. Dissolution of copper in sulphuric acid is also generally controlled by the proton attack and the generation of soluble cupric complexes and the further

involvement of nitric acid which provides new oxidative routes and may lead to a considerable increase in the proportion of corrosion rate (Guo *et al.* 2014). Traditional corrosion inhibitors likened to azoles, heterocyclic molecules as well as synthetic organics have been shown to be effective but their cost, toxicity and the inability to degrade are some of the factors that have prompted the exploration of new and safer options such as organophosphates (Mobin *et al.* 2018).

Green, environmentally-friendly as well as biodegradable inhibitors have gained popularity in recent years as a promising effective method of corrosion-mitigation (Lgaz *et al.* 2017). Particularly, attention has been directed towards expired pharmaceuticals since they also have heteroatoms (N, O, S), aromatic rings and conjugated structures that could adsorb to metal surfaces

in donor-acceptor interactions (Saha *et al.* 2016). Rather than wastage into chemical waste, the expired drugs can be used in the recycling of corrosion inhibitors to help towards sustainability and policies of waste-management (Fouda *et al.* 2017).

Adsorption of organic molecules onto copper metal surface typically pertains to either Langmuir- or Temkin-type of isotherms as well as physisorption, chemisorption or a combination there of. Amino acid functional groups (amides), imino acid functional groups (imines), carboxyl group (carboxylates), aromatic groups and sulfur groups can help scaffold surface coverage and minimise the rate of corrosion through protection film formation (Zhang *et al.* 2004). The performance of such molecules usually depends on concentration, temperature, acid strength, and molecular electronic attributes including HOMO-LUMO distribution.

Experiments with expired antibiotics, analgesics, antihistamines and antimalarial drugs indicate that they have high inhibition efficiencies on acidic media also supporting the feasibility of using pharmaceutical waste as a source of corrosion-control agents (Elmsellem *et al.* 2015). As such, the study of corrosion inhibition of copper in sulphuric and nitric acid environments presenting expired drugs has both scientific and environmental advantages. The objective of the current work is to examine the inhibition behavior, adsorption nature and protective coating of the chosen expired drug, which is informative of their possible application as sustainable corrosion inhibitors (Solmaz *et al.* 2014). The current work describes the corrosion prevention power of an outlived medication, Amisulpride, on the corrosion of copper in acidic media using the weight loss, Electrochemical Impedance Spectroscopy, Potentiodynamic polarization as well as, Quantum chemical investigations (Issaadi *et al.* 2011). Figure.1 provides the Structure of the drug molecule.

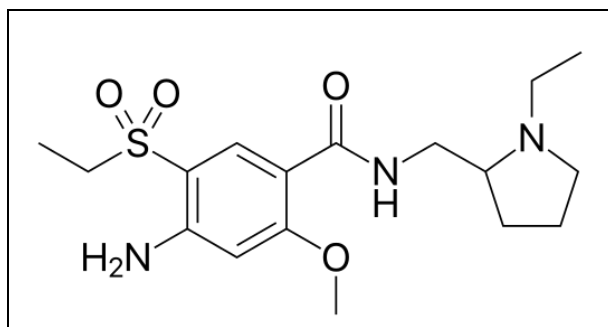


Figure .1 Structure of Amisulpride

2. MATERIALS AND METHODS

2.1 Composition of the Test Material

Copper specimen in the form of coupons of dimensions 2.5cm x 1cm x 0.1cm that were composed of

99.6 % Cu, 0.003 % Ni, 0.019 % Al, 0.002 % Mn and 0.116 wt.% Si were employed for weightless measurements. These specimens were honed with emery papers of 1/0, 2/0, 3/0, 4/0 as well as 5/0 grades and were de-greased using acetone. On the Teflon coated cylindrical copper electrode of exposed area 0.2826 cm² having the aforementioned composition, polarisation and impedance studies were carried out.

2.2 Preparation of the Stock Solutions

2.2.1 Preparation of acid solutions

The analytical reagent grade acids solutions were used to prepare bulk solutions of acids, which were diluted to required concentrations. All solutions including inhibitor stock solutions were prepared using double distilled water. The concentration of acid solutions were found using standardized NaOH solution.

2.2.2 Preparation of inhibitor solutions

The 2.5 M respective acid solutions were used to prepare the stock solutions of the inhibitors (100 mM) in 100 ml standard measuring flasks. Properly diluted bulk solution of the inhibitor was used to prepare test solutions which contained the given concentration of the inhibitor in the acids.

2.3 Instrumentation

Electrochemical experiments such as the potentiodynamic polarisation and electrochemical impedance measurements were conducted in a computer controlled Potentiostat (Zahner Zennium-An advanced Electrochemical Workstation) and the results were evaluated using a software, Zman Software. Electrochemical studies were performed using a three electrode configuration whereby, Pt foil was used as the auxiliary electrode, saturated calomel electrode (SCE) was used as the reference electrode and Teflon coated copper rod was the working electrode.

2.4 Experimental Techniques

2.4.1 Weight loss measurements

The weight loss measurements were taken using the American Society of Testing and Materials (ASTM) standard procedure [10]. The triplicate copper specimens were placed in 100 mL of the corrosive media with the inhibitor and without the inhibitor at a given temperature of 301 ± 1 K (room temperature). The mass of the two specimens before and after immersion was measured using a four decimal Electronic top loading balance which had a readability/sensitivity of 0.1mg and the average mass was used to derive the inhibition efficiency using the following formula [10]

$$IE\% = \frac{W - W'}{W} \times 100 \quad (1)$$

Where W and W' represent the weight loss in the uninhibited and inhibited solutions respectively.

2.4.2 Electrochemical impedance measurements

The stabilization delay before the process of impedance measurement could begin was taken to be 45 minutes after the immersion of the specimen in the corrosive media, whatever time was reasonable to ensure that $E_{ocp}/Vs(SCE)$ leveled off and reached a constant value. The measurements of the Electrochemical Impedance were conducted with the potentiostat (Zahner Zennium an Advanced Electrochemical Work Station). The experimental impedance spectra was recorded in the frequency range 100kHz to 10mHz, by superimposing a sinusoidal voltage of small amplitude 10mV. The inhibition efficiencies (IE%) were determined by the expression:

$$IE\% = \frac{R'_{ct} - R_{ct}}{R'_{ct}} \quad (2)$$

Where R_{ct} and R'_{ct} are the charge transfer resistance values in the uninhibited and inhibited solutions respectively.

2.4.3 Polarization measurements

The assessment of the effect of corrosion inhibition can be shown using various electrochemical methods and potentiodynamic polarization is one of the very potent methods. From this technique the current density of various potentials that give the information of the electrochemical behaviour and corrosion resistance of copper in the presence of the inhibitors can be obtained. The same cell set-up employed for EIS, was used to record potentiodynamic polarization curves at a potential sweep rate of $1.6mVs^{-1}$. The scan of potentials was made mainly in the negative end, than in the positive end of E_{ocp} , using corrosion potential (E_{corr}) and Tafel slopes β_a and β_c . The polarization curves of cathodic and anodic were drawn tangentially. The intersection of the two tangents i_{corr} and E_{corr} were estimated. The efficiencies of the inhibition were calculated using the relationship .

$$IE\% = \frac{i_{corr} - i'_{corr}}{i_{corr}} \times 100 \quad (3)$$

Where i_{corr} and i'_{corr} are the corrosion current densities in the absence and in the presence of inhibitor respectively.

2.4.4 Measurement of the potential of zero charge (PZC)

The physical adsorption is explained by the fact that the electrostatic force between the ions or dipole of the inhibiting species and the electrically charged surface of metal takes place. The surface charge of the metal is determined by the free corrosion potential (E_{corr}) and potential versus zero charge (PZC).

At 20 kHz AC frequency, the electrochemical impedance spectra were taken at different applied DC potentials. The results of the values of the double layer capacitance were plotted against applied DC potentials to obtain the potential of zero charge. The open circuit potential copper surface charge is estimated by using the equation .

$$E_r = E_{ocp} - E_{PZC} \quad (4)$$

Where E_r is antropov's rational corrosion potential, E_{ocp} is open circuit potential, E_{pzc} is potential at zero charge.

2.4.5 Adsorption isotherms and determination of thermodynamic parameters

Adsorption is a mathematical equation that connects bulk concentration of an adsorbing species with its surface concentration at a given temperature. An adsorption isotherm provides the dependence between interface coverage with the adsorbed species (i.e. amount adsorbed) and the concentration of species in solution. The appropriate isotherm may also be used to obtain the values of ΔG , ΔH , and ΔS of the adsorption of the inhibitor.

$$i_{corr} = Ae^{-E_a/RT} \quad (5)$$

Where i_{corr} is corrosion current density, A is the Arrhenius pre-exponential constant, R is universal gas constant, E_a is the energy of activation and T is the absolute temperature.

2.4.7 Quantum chemical calculations

Complete geometry optimization was implemented all in quantum chemical calculations with code Gaussian 98. GaussView 3.0 is used to sketch the molecular structures of the inhibitor. The density functional theory (DFT) level with the non-local hybrid density functional B3LYP, that is the combination of the three-parameter hybrid exchange functional by Becke and the correlation functional by Lee *et al*, at the basis sets 3-21G and 6-31G(d,p) was used to achieve geometry optimizations . The geometry was optimized and a vibrational analysis was used followed by verification of the geometry against actual minima on the potential

energy surface as indicated by the non-existence of imaginary frequencies.

3. RESULTS AND DISCUSSION

3.1 Weightloss Measurements

Table 1 shows the values of inhibition efficiency of copper in 0.5 M solution of H₂SO₄ and 1.0 M solution of HNO₃ acid at various concentrations of Amisulpride. It is observed that the more the concentration of the inhibitor, the more the efficiency of inhibition. This is because of the high affinity of the Amisulpride drug molecules with the metal surface leading to adsorption. The degree of adsorption rises as the concentration of the inhibitor is increased to a higher degree resulting in a higher degree of efficiency and the optimum inhibition efficiency was at 1 mM of the inhibitor concentration (Feng *et al.* 2019). Organic inhibitors have the general effect of refuting the dissolution of the metal by forming a protective coating on the metal surface and isolating the metal and the medium of corrosion (Keles *et al.* 2008). The inhibitory ability of the inhibitor molecule is a result of the propensity to establish a strong or weak chemical bond with the Fe atoms with the aid of the lone pair of electrons that is present on the O, S and N atom and the electrons of >C=O-, S=O moieties and benzene ring (Li *et al.* 2009).

Table 1 Inhibitor efficiency from weight loss measurement for copper in 0.5 M H₂SO₄ and 1.0 M HNO₃ solutions in the presence and absence of Amisulpride.

Medium	Inhibitor Concentration (mM)	Corrosion rate (mpy)	Inhibitor Efficiency (IE%)
0.5 M H ₂ SO ₄	Blank	23.72	-
	0.05	11.12	53.1
	0.1	8.25	65.2
	0.2	5.12	78.4
	0.5	2.11	91.1
	1.0	1.12	95.3
	2.0	1.12	95.3
1.0 M HNO ₃	Blank	48.11	-
	0.05	24.92	48.2
	0.1	17.12	64.4
	0.2	11.88	75.3
	0.5	8.61	82.1
	1.0	5.62	88.3
	2.0	5.62	88.3

3.2 Electrochemical Impedance Spectroscopic Studies

Figure 2 and 3 respectively displayed the impedance spectra of copper corrosion at room temperature in 0.5 M H₂SO₄ solution and 1.0 M HNO₃ solutions in the presence and absence of Amisulpride. The Nyquist plots are just semi circles without inductive loop etc. A simple model, which is -R(CR)- more appropriately elucidates the case at the metal/solution interface. The semicircles are depressed which indicates the non-ideal capacitance of the electrochemical solid/liquid interface ((Kadiri *et al.* 2025)). The capacitance dispersion at the surfaces of the solids was possibly caused by the roughness of the surface, the polymorphism of the polycrystalline, the chemical heterogeneities and adsorption of the anions. The equivalent circuit (Figure 4) is so accurately modelled by substituting the capacitor (capacitive element) by a constant phase element (cpe) and the impedance functionality of which is given by the expression;

$$Z_{cpe} = \frac{1}{Y_0(j\omega)^n} \quad (6)$$

Where Y_0 is the admittance of the corrosive system at 1 rad s^{-1} and n is a constant, ($-1 < n < 1$). When $n = 0$, the cpe represents a pure resistor, if $n = +1$ the cpe represents a pure capacitor and if $n = -1$ it represents an inductor.

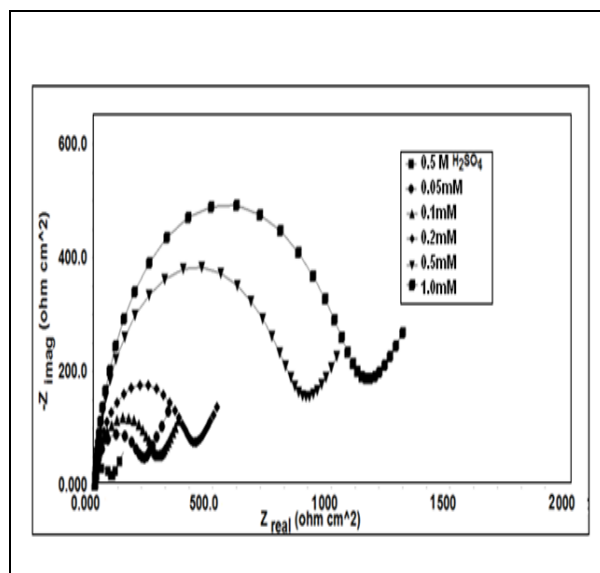


Figure.2 The Nyquist plots for Cu electrode obtained in 0.5 M H₂SO₄ solutions in the absence and presence of Amisulpride at various concentrations.

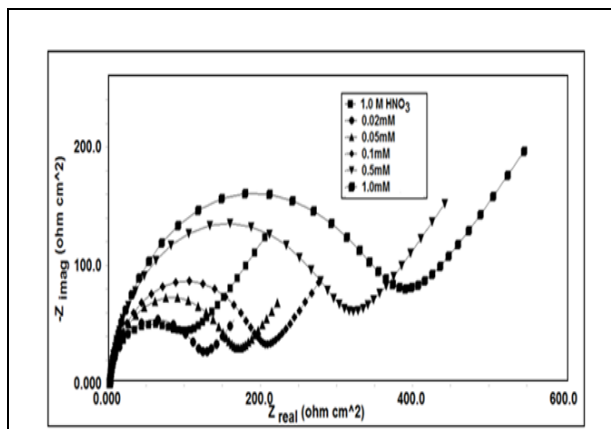


Figure.3 The Nyquist plots for Cu electrode obtained in 1.0 M HNO₃ solutions in the absence and presence of Amisulpride at various concentrations.

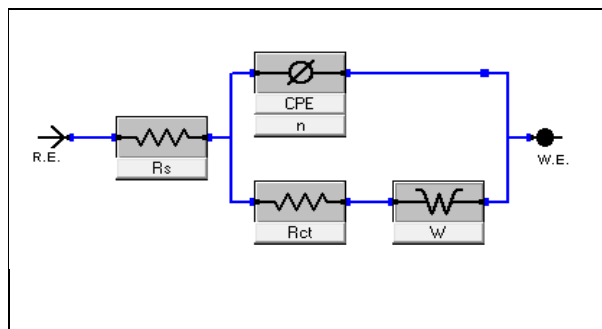


Figure.4 An Equivalent Circuit for copper metal surface - solution interface.

By using the proposed equivalent circuit provided in Figure 4, the electrochemical impedance parameters of copper corrosion in 0.5 M H₂SO₄ and 1.0 M HNO₃ solutions in the presence and absence of Amisulpride are obtained by the fitting of the experimental impedance curves. Table 2 gives these impedance parameters.

Figure.4 Table 2 Electrochemical impedance parameters for copper in 0.5M H₂SO₄ and 1.0 M HNO₃ solutions in the presence and absence of Amisulpride from EIS.

Medium	Inhibitor Concentration (mM)	R_{ct} Ωcm^2	Y_0 ($\times 10^{-6}$) $\Omega^{-1}\text{cm}^{-2}$	n	C_{dl} μFcm^{-2}	Inhibitor Efficiency (IE %)
0.5M H ₂ SO ₄	Blank	61.22	451.3	0.911	155.2	-
	0.02	270	245	0.926	119	77.3
	0.05	338	192	0.932	95	81.9
	0.10	518	157	0.922	72	88.2
	0.50	1007	105	0.961	42	93.9
	1.0	1198	95	0.967	31	94.9
	Blank	55.20	618.4	0.905	168.4	-
1.0 M HNO ₃	0.02	178	304	0.925	125	68.9
	0.05	205	247	0.918	112	73.1
	0.10	235	188	0.927	101	76.5
	0.50	347	123	0.939	94	84.1
	1.0	421	107	0.968	78	86.9
	Blank	55.20	618.4	0.905	168.4	-

As observed in Table 2, the R_{ct} value rose and C_{dl} value fell as the inhibitor concentration increased in 0.5 M H₂SO₄ and in 1.0 M HNO₃ solutions. The increase in the R_{ct} values with the increase in the inhibitor concentration is attributed to the increase in the surface coverage with the Amisulpride drug molecules, which caused the increase in the inhibitor efficiency (Xu *et al.* 2013). The decrease of the C_{dl} values with the rise of the inhibitor concentration is due to the adsorption of the inhibitor molecules into the interface of the metal

solution leading to the decrease of the local dielectric constant and/or an increase in the thickness of the electrical double layer (Xu *et al.* 2014). Therefore, the reduction in the C_{dl} values due to the slow replacement of the water molecules by the adsorption of the inhibitor molecules at the metal solution interface implies reduced corrosion rate as the presence of the inhibitor molecules is felt.

3.3 Potentiodynamic Polarization Studies

Figure 5 and 6 represent the Tafel plot of corrosion of copper in 1.0 M nitric acid and 0.5 M sulphuric acid solution in the absence and presence of Amisulpride respectively.

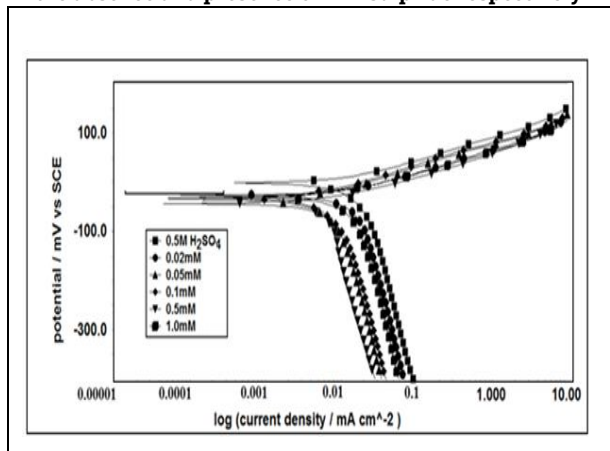


Figure.5 Tafel plots for copper electrode in 0.5 M H₂SO₄ solutions in the absence and presence of the Amisulpride at various concentrations.

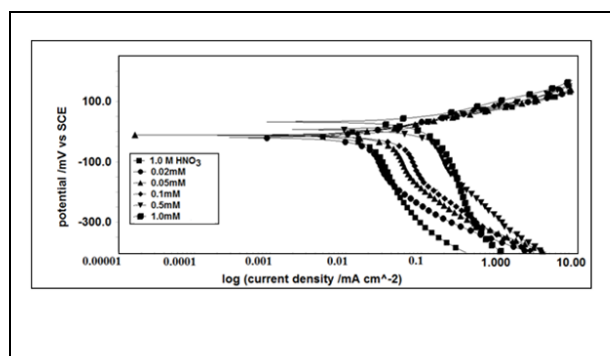


Figure.6 Tafel plots for copper electrode in 1.0 M HNO₃ solutions in the absence and presence of the Amisulpride at various concentrations.

Based on the Tafel plots, it is observed that cathodic Tafel slope vary after added Amisulpride and anodic Tafel slope doesn't show significant variation. This indicates that the Amisulpride molecules show much effects on cathodic reaction than anodic reaction (Qiang *et al.* 2018). The inhibitor is, therefore, classified as cathodic. This shows that activation control is applied in the cathodic reaction. The absence of steep lines in the anodic area shows that no passive film is formed on copper face.

Table 3 presents the electrochemical parameters consisting of corrosion potential (E_{corr}), corrosion current density (i_{corr}) and cathodic and anodic slopes (β_c and β_a), surface coverage and the inhibitor efficiency. The value of β_c is higher than β_a . This meant that the inhibitor molecules slow down the cathodic reaction more than anodic reaction. This is also credited to the fact that the rate at which the hydrogen evolution reaction occurs on copper metal surface is reduced due to adsorptions of the inhibitor molecules to metal surface.

Table 3 Electrochemical parameters for copper in 0.5 M H₂SO₄ and 1.0 M HNO₃ solutions in the presence and absence of Amisulpride from Tafel polarization curves.

Medium	Inhibitor Concentration(mM)	E_{corr} (mV)	I_{corr} ($\mu A cm^{-2}$)	β_c	β_a	Inhibitor Efficiency (IE %)
0.5 M H ₂ SO ₄	Blank	25	45.81	86	69	-
	0.02	-28	32.5	381	51	29
	0.05	-32	21.9	389	48	52
	0.1	-45	12.4	544	46	73
	0.5	-35	9.2	244	51	80
	1.0	-32	7.4	205	49	84
	Blank	66	371.3	89	68	-
1.0 M HNO ₃	0.02	7.4	278.5	204.0	72	25

Medium	Inhibitor Concentration(mM)	E_{corr} (mV)	I_{corr} ($\mu A cm^{-2}$)	β_c	β_a	Inhibitor Efficiency (IE %)
1.0M HNO ₃	0.05	-15.2	137.4	161.5	69	63
	0.1	-12.3	118.8	147.7	55	68
	0.5	-13.5	92.8	265.6	49	75
	1.0	-21.4	81.7	136.6	51	78

3.4 Adsorption Isotherm

Experimental findings on the effects on corrosion of copper metal in 1.0M nitric acid and 0.5 M sulphuric acids revealed that the Amrisulpride adsorption process on the copper metal surface was more likely to follow Langmuir adsorption isotherm (Vasziicsin *et al.* 2023).. The graphical representation of either of $C_{inh}/C_{inh} + \theta$ was observed to be a straight-line in both the media with the correlation coefficient being more than 0.9. Figure 7 shows the Langmuir adsorption isotherm of adsorption of Amrisulpride on the surface of copper metal.

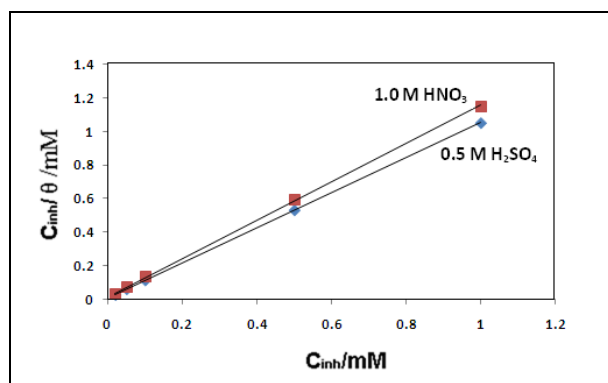


Figure 7 Langmuir adsorption Isotherm plots for adsorption of Amrisulpride onto copper surface in 0.5 M H₂SO₄ and 1.0 M HNO₃ acid solutions.

Table 4 shows the parameters of adsorption kinetics. Based on this table it can be seen that standard free energy change (ΔG^0) of adsorption of Amrisulpride on the copper metal surface is negative in both the acid media, which implies spontaneous adsorption of

Amisulpride. Free energy of adsorption values normal standard of 1.0M nitric acid and 0.5 M sulphuric acid solution of Amisulpride were -23.00 and -21.81 kJ mole⁻¹ respectively. These values are near -20 kJ mole⁻¹ and therefore, adsorption process is mainly physisorption. Nevertheless, the values are a little larger than -20 kJ mole⁻¹, which demonstrates a minor role played by chemisorption.

Table.4 Thermodynamic parameters obtained from adsorption isotherm.

Medium	$K^{ads}(M^{-1})$	$\Delta G^0(kJ/mole)$	Correlation coefficient (R)
0.5M H ₂ SO ₄	149.5	-23.00	1
1.0M HNO ₃	95.25	-21.81	0.999

3.5 Effect of Temperature

To determine the value of the activation energy as well as an understanding of the mechanism used in removing the inhibitors adsorbed in the surface of copper, potentiodynamic polarization study was conducted at 308, 318 and 328 K. Figures 8, 9, 10 and 11 have the Tafel plots obtained at 308, 318 and 328 K in the presence and absence of Amrisulpride.

The value of corrosion current density calculated by Tafel plots were indicated in Table 5 (in the absence and presence of Amrisulpride at different temperatures).

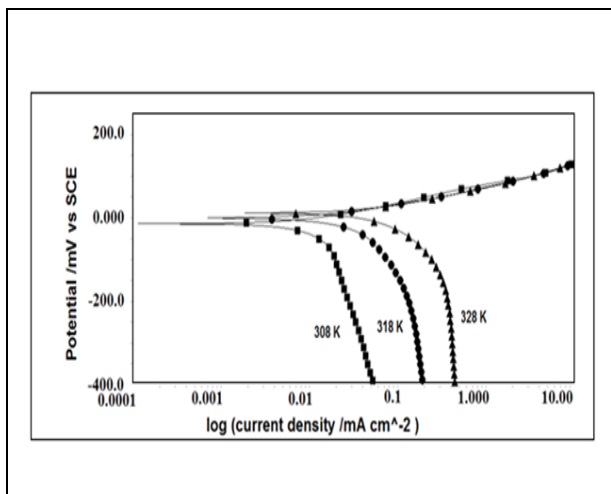


Figure.8 Tafel plots for corrosion of copper in 0.5 M H_2SO_4 acid solution at 308, 318 and 328 K.

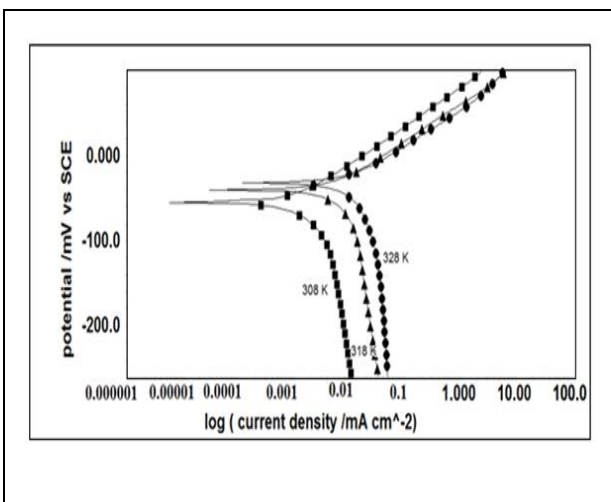


Figure.9 Tafel plots for corrosion of copper in 0.5 M H_2SO_4 acid solution in the presence of 1.0 mM Amisulpride at 308, 318 and 328 K.

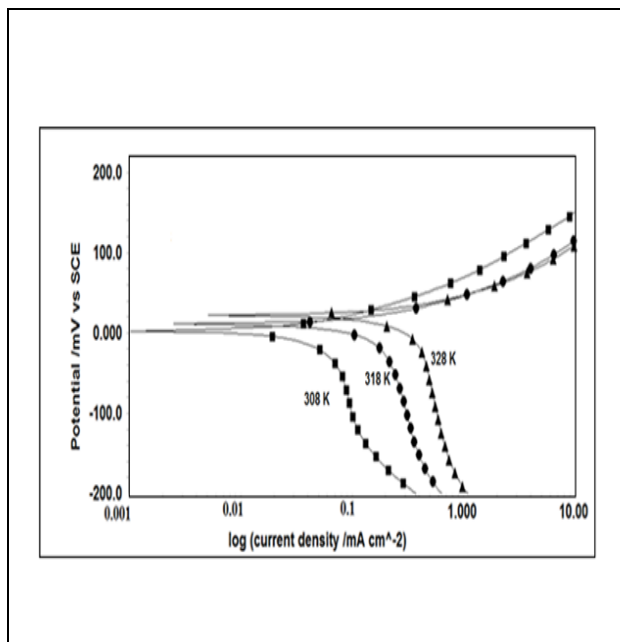


Figure.10 Tafel plots for corrosion of copper in 1.0 M HNO_3 acid solution at 308, 318 and 328 K.

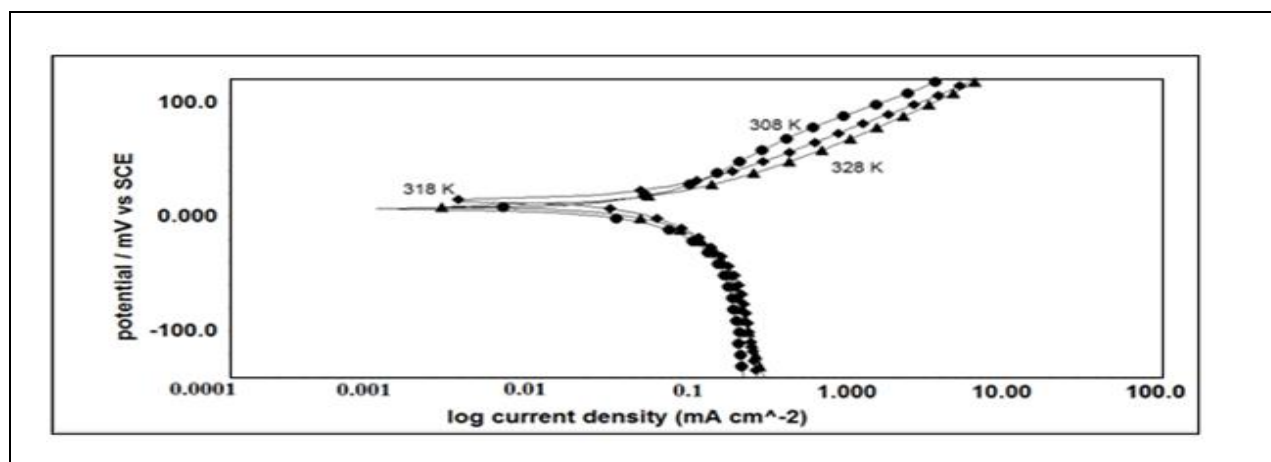


Figure.11 Tafel plots for corrosion of copper in 1.0 M HNO₃ acid solution in the presence of 1.0 mM Amisulpride at 308, 318 and 328 K.

Table 5 Corrosion current density values for corrosion of copper in the absence and presence of Amrisulpride in 0.5 M H₂SO₄ and 1.0 M HNO₃ solutions at 308,318 and 328 K

Medium	Concentration of Inhibitor (mM)	Temperature (K)	i_{corr} (μA)
0.5 M H ₂ SO ₄	Blank	308	12.4
	Blank	318	31.1
	Blank	328	89.1
	1.0mM	308	7.1
	1.0mM	318	12.5
	1.0mM	328	16.3
1.0 M HNO ₃	Blank	308	30.5
	Blank	318	198.7
	Blank	328	361.2
	1.0mM	308	14.9
	1.0mM	318	73.1
	1.0mM	328	94.2

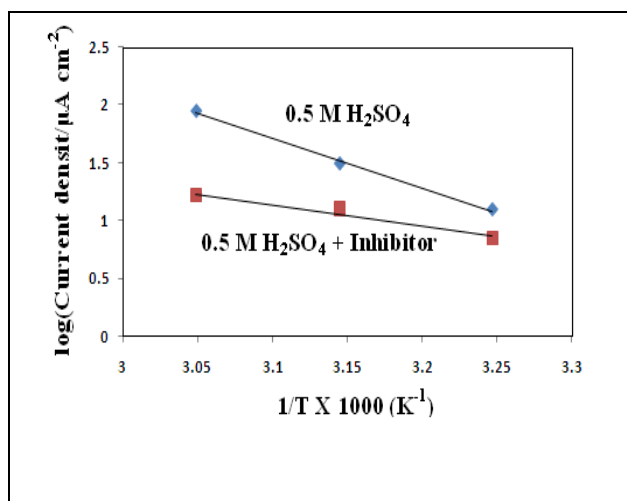


Figure.12 Arrhenius plots for corrosion of copper in 0.5 M H₂SO₄ acid solution in the presence and absence of Amrisulpride at 308,318 and 328 K

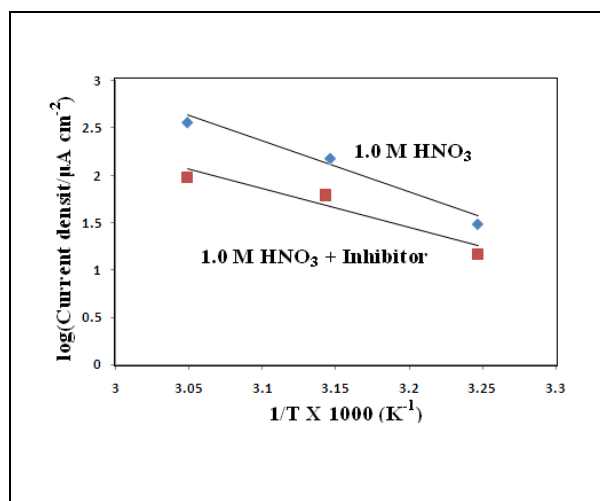


Figure.13 Arrhenius plots for corrosion of copper in 1.0 M HNO₃ acid solution in the presence and absence of Amrisulpride at 308,318 and 328 K

The energy of activation (E_a) values calculated from Arrhenius plots are presented in Table 6.

Table.6 Energy of Activation for the corrosion of copper in 0.5 M H₂SO₄ and 1.0 M HNO₃ solutions in the absence and presence of Amrisulpride

Medium	Inhibitor concentration(mM)	E_a (kJ / mole)
0.5 M H ₂ SO ₄	Blank	61.39
	1mM Amrisulpride	57.11
1.0 M HNO ₃	Blank	68.22
	1mM Amrisulpride	33.48

It can be observed in the table above that the values of E_a decreased with respect to blank solution in the presence of Amrisulpride. This is attributed to the fact that the energy barrier to the adsorption of the Amrisulpride molecules on copper surface reduces as temperatures rises which leads to the growth of the the Amrisulpride molecules on Cu surface with an increase in temperature. Diffusion of the product of the corrosion process through the protective film deposited on the metal surface by the inhibitor molecules controls the rate of copper dissolution in the corrosive media . It shows that the molecules of the inhibitor adsorbed on the copper metal are by chemisorption to the surface of the metal when E_a increases with the addition of the Amrisulpride to both the media .

3.6 Potential of Zero Charge

Figure 14,15, 16 and 17 show the dependence of the change in capacitance between the two metal layers on applied DC potential with an AC frequency of 100 Hz in absence and presence of Amisulpride on the copper metal - solution interface in the presence of the two solutions (0.5 M H₂SO₄ and 1.0 M HNO₃). Open circuit potential and PZC values obtained as a result of these plots are shown in Table 7.

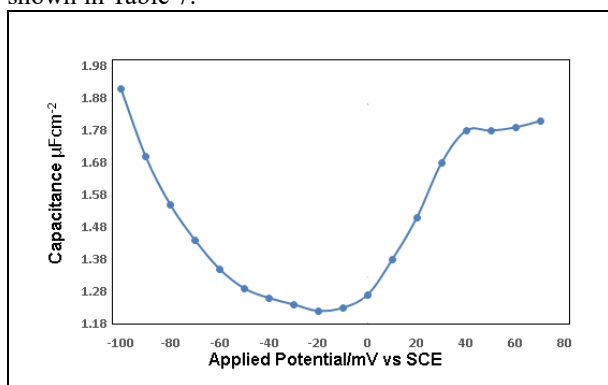


Figure.14 Plot of differential capacitance vs. applied electrode potential for copper in 0.5 M H₂SO₄.

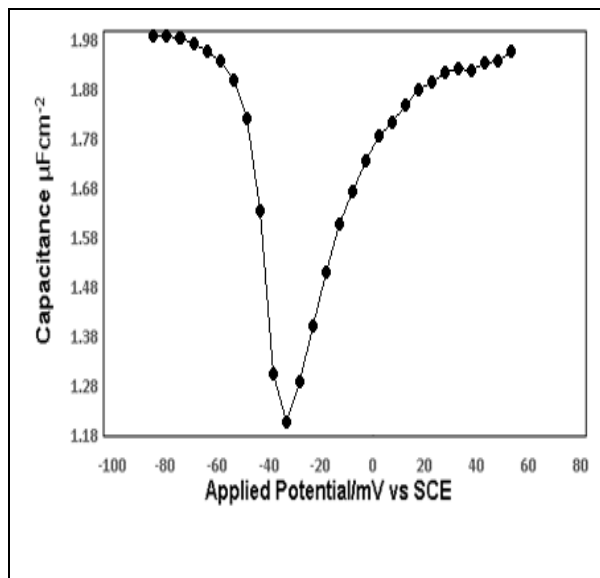


Figure.15 Plot of differential capacitance vs. applied electrode potential for copper in 0.5 M H₂SO₄ containing 1.0 mM Amrisulpride.

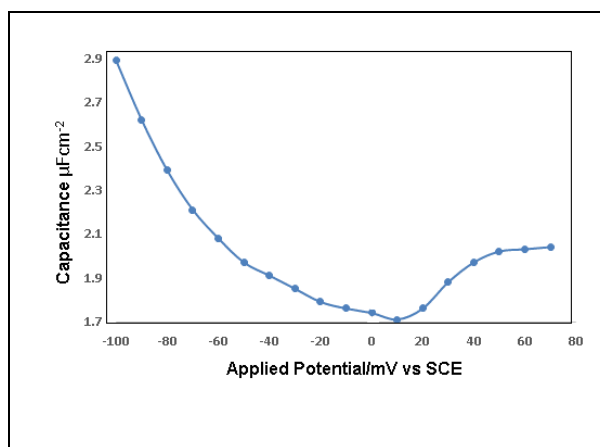


Figure.16 Plot of differential capacitance vs. applied electrode potential for copper in 0.5 M HNO₃.

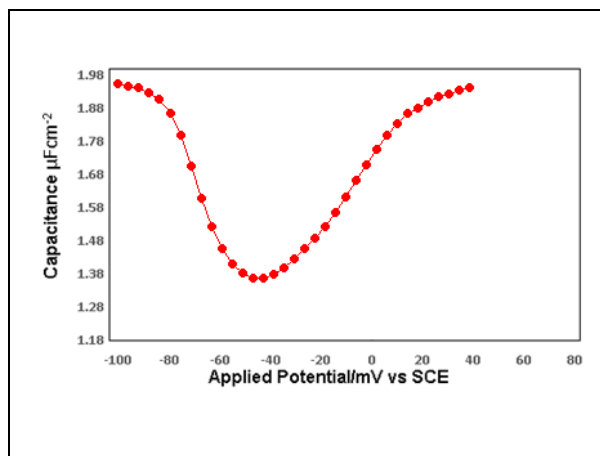


Figure.17 Plot of differential capacitance vs. applied electrode potential for copper in 0.5 M HNO₃ containing 1.0 mM Amrisulpride.

Table .7 Excess charge on Cu metal surface in 0.5M H₂SO₄ and 1.0M HNO₃ solutions in the presence and absence of the Amrisulpride.

Medium	EOCP (mV/SCE)	PZC (mV/SCE)	Excess charge (EOCP- EPZC)
1.0 M HNO ₃	66	10	+56
1.0 M HNO ₃ +1.0 mM Amrisulpride	104	-37	+141
0.5M H ₂ SO ₄	25	-20	+45
0.5M H ₂ SO ₄ +1.0 mM Amrisulpride	153	-52	+205

Based on Table 7 it is apparent that copper metal surface is positively charged in the acid media and based on that, the following mechanism was postulated to explain corrosion of copper in the acid media. Without an inhibitor, successively reversible adsorption of anions ($\{SO\}_4^{-2}$ and $\{NO\}_3^-$) on the copper surface, desorption of electrons of the adsorbed anion on the copper surface and desorption of the adsorbed anion and Cu^{+2} ions, with electrons being picked off of the copper atoms, follow. The Amisulpride molecules in the inhibited solutions may be protonated and will exist in complex equilibria between various protonated species with species of neutral (unprotonated) molecule. The protonated molecules are positively charged and therefore the approaching molecules cannot directly go to the positively charged metal surface in acidic solutions.

Nonetheless, they are capable of attaching themselves on copper metal surface via sulphate and nitrate anions bridges. Besides this form of physical interaction, the unprotonated molecules may also adsorb onto surface Cu atoms by donating lone pair of electrons on N,O and S atoms as well as the p electrons of S=O group and benzene ring to empty d orbitals of the surface atoms .

3.7 Molecular Structure and Quantum Chemical Calculations

To define the performance of an inhibitor, it is effective to employ spatial distributions of atoms and groups and electronic structures of an inhibitor. Compared to natural sources of drugs, totally laboratory prepared drugs are achiral or are, with the exception of achiral ones, typically prepared and marketed in racemic mixtures. Racemic Amisulpride has both (+) and (-) enantiomers in equal concentration, and both of them are mixed in solution; therefore, both are present in quantum chemical calculations. Figure 8.18, depicts the stereo image of the optimized pair of enantiomers of amisulpride racemic molecule. It is a fact that two non-superimposable, mirror-image stereo isomers have the same physical properties except that one can spin around them clockwise or anticlockwise plane-polarized light.

Looking at the above argument, and owing to the total coincidence between the structural and electronic properties of two enantiomers obtained (Figure 18, Table 8 and Table 9), research will be provided below on one of the enantiomers whose outcomes can be overlaid onto the other. Because the copper is already reacted with the inhibitor in acidic pH, involvement of protonated structures of Amisulpride, other than the homogeneous neutral form, have been anticipated to be involved in the corrosion inhibition process.

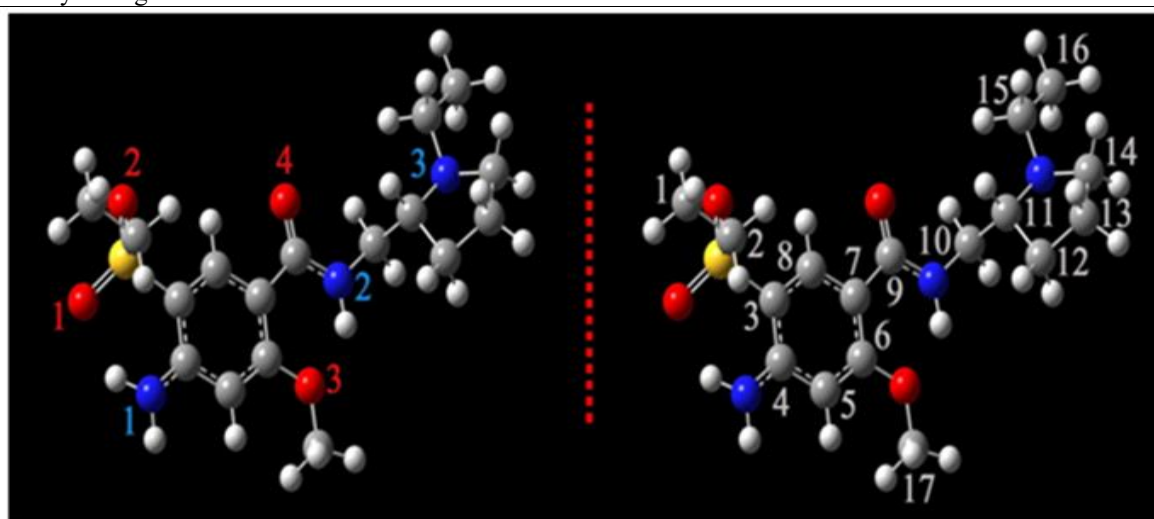


Figure.18 Stereo view of Optimized Structures of pair of Amisulpride.

Table.8 Selected Bond Lengths (in angstroms) and Bond Angles (in degrees) of the (A) Amisulpride, (B) (Amisulpride)N3H⁺, (C) (Amisulpride)O4H⁺, (D) (Amisulpride)O2H⁺, (E) (Amisulpride)O1H⁺, (F) (Amisulpride)N3O2H2²⁺, (G) (Amisulpride)O1O4H2²⁺, and (H) (Amisulpride)O2O4H2²⁺ structures using the B3LYP/6-31G(d,p) method.

	A	B	C	D	E	F	G	H
Bond Lengths								
S-O1	1.48500	1.48216	1.47883	1.46776	1.63920	1.46385	1.62394	1.46015
S-O2	1.47001	1.46986	1.47019	1.62833	1.45671	1.62117	1.45669	1.61535
C4-N1	1.37498	1.36194	1.34795	1.35150	1.38003	1.34586	1.35476	1.34355
C6-O3	1.36745	1.36600	1.35339	1.34547	1.34481	1.34726	1.33595	1.33704
C7-C9	1.50985	1.48274	1.44787	1.52348	1.52638	1.50372	1.46914	1.46647
C9-O4	1.23218	1.24023	1.33024	1.23202	1.22967	1.23520	1.32823	1.33073
C9-N2	1.36221	1.37439	1.31469	1.34997	1.35112	1.36421	1.30588	1.30554
N2-C10	1.45512	1.44337	1.47342	1.46145	1.46194	1.45000	1.47754	1.47761
C11-N3	1.47153	1.54596	1.46084	1.46587	1.46619	1.54372	1.44977	1.45043
C14-N3	1.47818	1.53257	1.48322	1.48035	1.48014	1.53896	1.48337	1.48355
C15-N3	1.45975	1.52288	1.46397	1.46111	1.46147	1.52144	1.46590	1.46584
Avg. of (C-C)Ar	1.40402	1.40535	1.40850	1.40900	1.40648	1.41019	1.41110	1.41286
Bond Angles								
C1-C2-S	110.82569	110.64885	110.43657	111.47013	112.24589	111.73271	111.77484	111.84519
O1-S-O2	119.61601	120.27113	120.91480	111.80224	112.99413	113.17094	114.22503	114.58966
C6-O3-C17	119.79074	119.62738	120.03018	120.56337	120.82693	120.34244	120.95041	120.71511
C7-C9-N2	118.15586	119.34782	124.44877	118.02738	118.12691	118.86576	124.08755	124.16682
N3-C15-C16	112.41940	111.95875	112.48507	112.42940	112.44398	112.10477	112.62951	112.62389
C1-C2-S-C3	-177.79027	-176.52643	-177.24077	174.42914	-175.05138	174.58619	-175.85213	176.05068
C7-C9-N2-C10	-178.94057	172.02062	179.36390	-179.23654	-179.22692	171.26476	-177.67625	-178.95906
C9-N2-C10-C11	87.80945	74.96915	93.84147	101.49306	96.71279	78.83189	95.50726	96.92695
N2-C10-C11-N3	-177.84485	-143.14499	179.38634	-179.97501	-178.49605	-156.95812	179.33447	179.08048
C10-C11-N3-C15	79.40665	90.21326	83.59933	79.72295	80.57172	89.83375	86.17007	86.25031
C11-N3-C15-C16	-160.36266	-163.01666	-159.64315	-160.93217	-160.16249	-162.92593	-157.03482	-156.89913

Table.9 Some molecular properties of (A) Amisulpride, (B) (Amisulpride)N3H⁺, (C) (Amisulpride)O4H⁺, (D) (Amisulpride)O2H⁺, (E) (Amisulpride)O1H⁺, (F) (Amisulpride)N3O2H2²⁺, (G) (Amisulpride)O2O4H2²⁺, and (H) (Amisulpride)O1O4H2²⁺ structures calculated using DFT at B3LYP/6-31G(d,p) basis set.

Quantum chemical properties	A	B	C	D	E	F	G	H
Total energy (eV)	-41563.76	-41574.83	-41574.29	-41573.32	-41573.30	-41582.46	-41580.88	-41580.94
ELUMO (eV)	-0.783	-3.413	-5.303	-4.982	-5.122	-6.997	-8.475	-8.443
EHOMO (eV)	-5.355	-8.287	-8.075	-7.216	-7.150	-11.689	-9.986	-9.964
ΔE (eV)	4.572	4.875	2.773	2.234	2.028	4.693	1.511	1.521
dipole moment (D)	8.801	17.599	7.386	14.346	17.516	4.900	13.856	16.270
I (eV)	5.355	8.287	8.075	7.216	7.150	11.689	9.986	9.964
A (eV)	0.783	3.413	5.303	4.982	5.122	6.997	8.475	8.443
η (eV)	2.286	2.437	1.386	1.117	1.014	2.346	0.756	0.761
S (eV ⁻¹)	0.219	0.205	0.361	0.448	0.493	0.213	0.662	0.657
χ (eV)	3.069	5.850	6.689	6.099	6.136	9.343	9.231	9.204
ΔN	-0.379	0.215	0.681	0.581	0.659	0.968	2.932	2.895
ω (eV)	2.060	7.020	16.138	16.651	18.565	18.602	56.376	55.678
ΔE (eV)	-0.571	-0.609	-0.347	-0.279	-0.254	-0.587	-0.189	-0.190
molar volume (cm ³ mol ⁻¹)	293.964	269.855	269.046	287.547	274.000	260.486	292.186	280.211

Addition of protons to Amisulpride molecule at various positions has been done in the DFT calculation. Though the analysis of the natural bond orbital (NBO) charge population of Amisulpride in Figure 19 indicates that both oxygen atoms and nitrogen atoms bear excess negative atomic charge, the 3D plot of the molecular electrostatic potential map (MESP) surface of this molecule (Figure 20) indicates that there is a potentiality of having four protons to the molecule. The MESP holds data on the complete electronic constitute of the molecule and outlines nucleophilicity as well as electrophilicity areas on the molecular surface. Any proton that comes

closer will first of all be drawn to those areas where the potential are negative values. The four large negative potential areas near oxygen atom of the carbonyl group (O4) (reflected by red color), near two oxygen atom of the sulfonyl group (O1 and O2) (reflected by red color) and near nitrogen atom of the pyrrolidine ring (N1) illustrated by yellow color are clearly seen in the MESP of Amisulpride. The 2D MESP contour maps in Figure 19 give all the information related to the distribution of MESP and it can be outlined by visualizing it as a manifold of the spatial position with respect to the molecule.

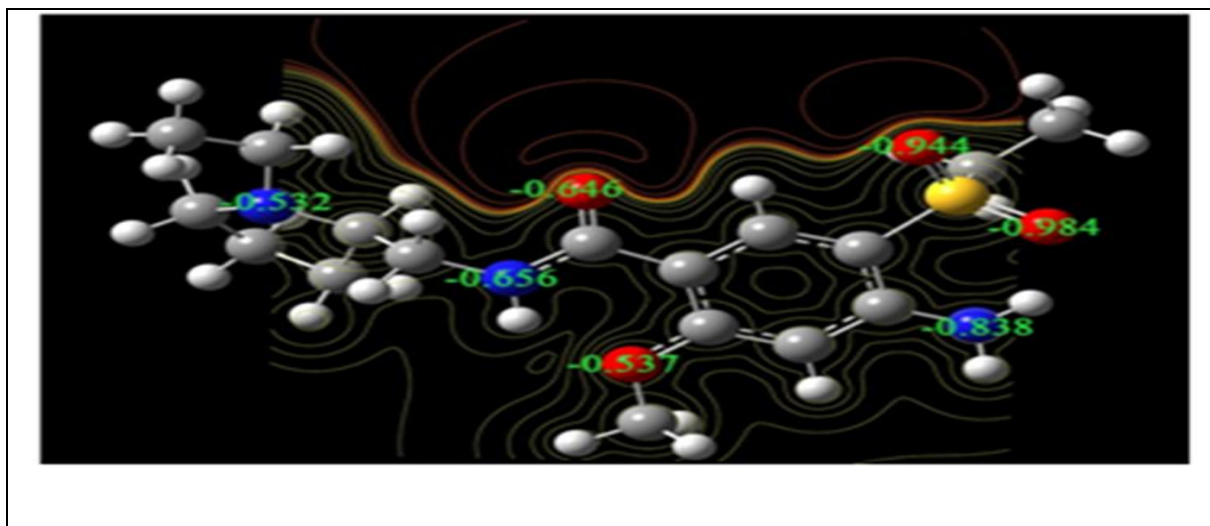


Figure.19 NBO charge population in Amisulpride.

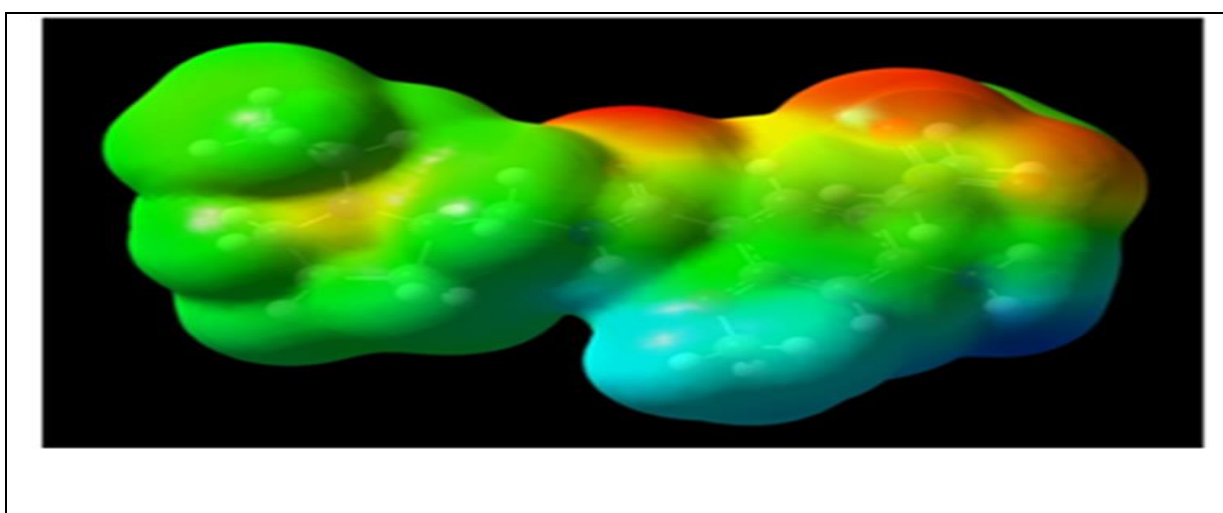


Figure.20 Molecular electrostatic potential map surface of Amisulpride.

The structural and electronic properties of Amisulpride, and the protonation structures of Amisulpride have been studied, and results have indicated that there is a significant impact on the chemical properties, as that of the electron-donating ability to the metal.

The bond distances of Amisulpride were calculated and the result is in excellent agreement with the experiment values (Figure 21 and Table 8). To illustrate, the average length of the chemical bonds between the two atoms C9-O4 and C9-N2 are 1.23 and 1.36 respectively in the neutral form of the molecule, which means that there exists the localization of the double bonds, with the C9 center being sp² hybridized. Moreover, S-O1 and S-O2 bonds are nearly equal in length and approximately equal at 1.47-1.48 Å and indicate that both oxygen atoms (O1 and O2) are in sp² hybridization. These structural parameters indicate that the molecule can be relatively electron-dense, that is,

there is some electron density that can be donated to a different species. Conversely, when protonation is done

on O4, the region (O4=C9=N2) becomes delocalized to obtain a symmetrical structure. Nevertheless, the protonation of either O1 or O2 leads to the eventuation of breaking the symmetry of the region (O1=S=O2) (Figure 20 and Table 8).

C1-C2-S bond angle, C6-O3-C17 bond angle and N3-C15-C16 bond angle are close in all the geometries (Table 8). Nevertheless, (C9-N2-C10-C11= 74.97 and 78.83) of (Amisulpride)N3H⁺ and (Amisulpride)N3O2H2+2 structures are less dihedral compared to the other structures, suggesting that these two structures are more planar, than the other structures. That way, one can conclude that the more planarity are achieved in protonated structures than the neutral one does, because protonated ones can be corrosion inhibited more than the neutral form.

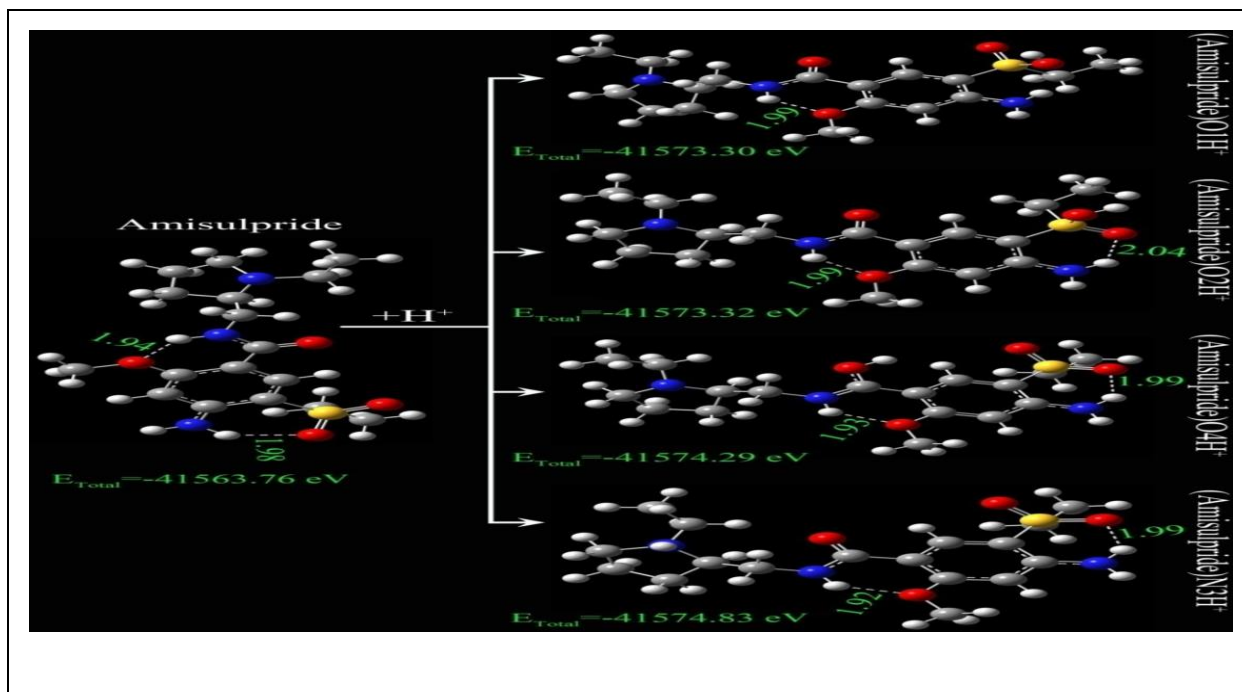


Figure.21 Optimized structures of various protonated forms of Amisulpride

The theoretical data indicate that in the case of Amisulpride (Figure 22 and 23) a higher proportion of stable structures were determined at each of the four of the above-mentioned probable regions, especially at a nitrogen (N_3); when the second proton was added to the molecule, especially at O_2 , a total energy of each of the molecules dropped significantly and, in this case, (Amisulpride) $N_3O_2H_2+2$ of -41582.46 eV. As Figure 23 and 24 reveal, it was easy to notice that various structures of amisulpride molecule are not entirely planar, which might lead to rather weak interaction between molecules and metal surface. It can be observed that unlike the other

structures, (Amisulpride) N_3H^+ structure of amisulpride is rather planar. Nevertheless, there are other variables, which should be taken into consideration to explain the orientation of organic characters within the electrode surface. The nanotechnical structure of the electrode surface can be affected by the geometry of the inhibitor and the nature of the frontier molecular orbitals of the atoms and groups of the molecules. Frontier molecular orbital (HOMO and LUMO) theory comes in handy when forecasting the adsorption centers of the inhibitor that results in the interaction with metal surface.

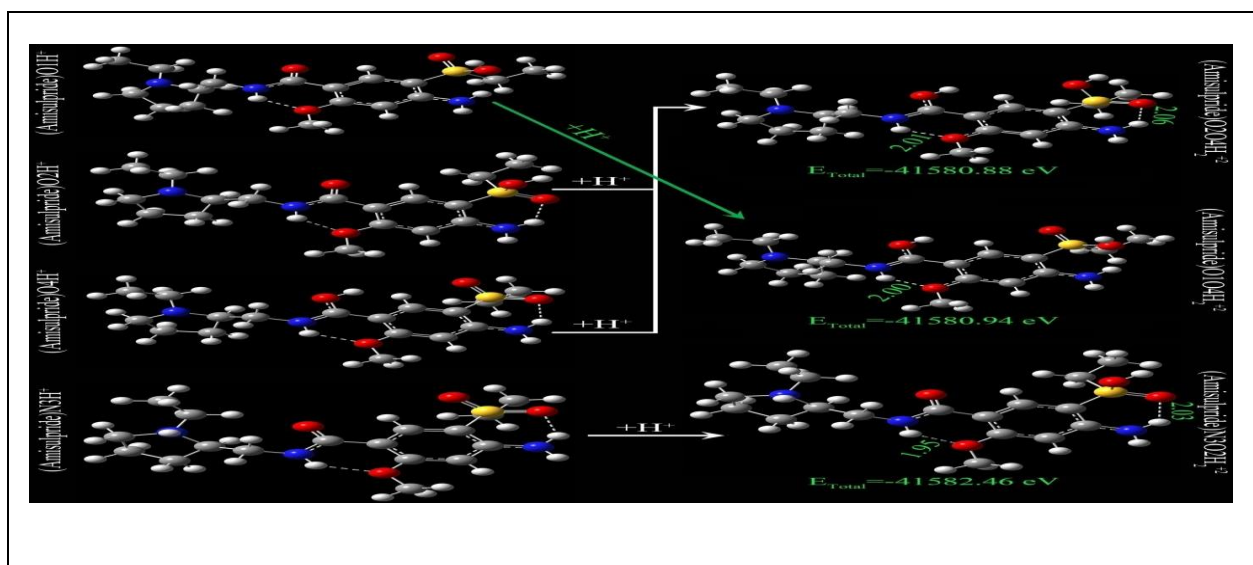


Figure.22 Optimized structures of come protonated forms of Amisulpride and their energies.

The HOMO and LUMO populations of the studied amisulpride molecule and the corresponding protonated forms are shown in Figure 23 and 24.

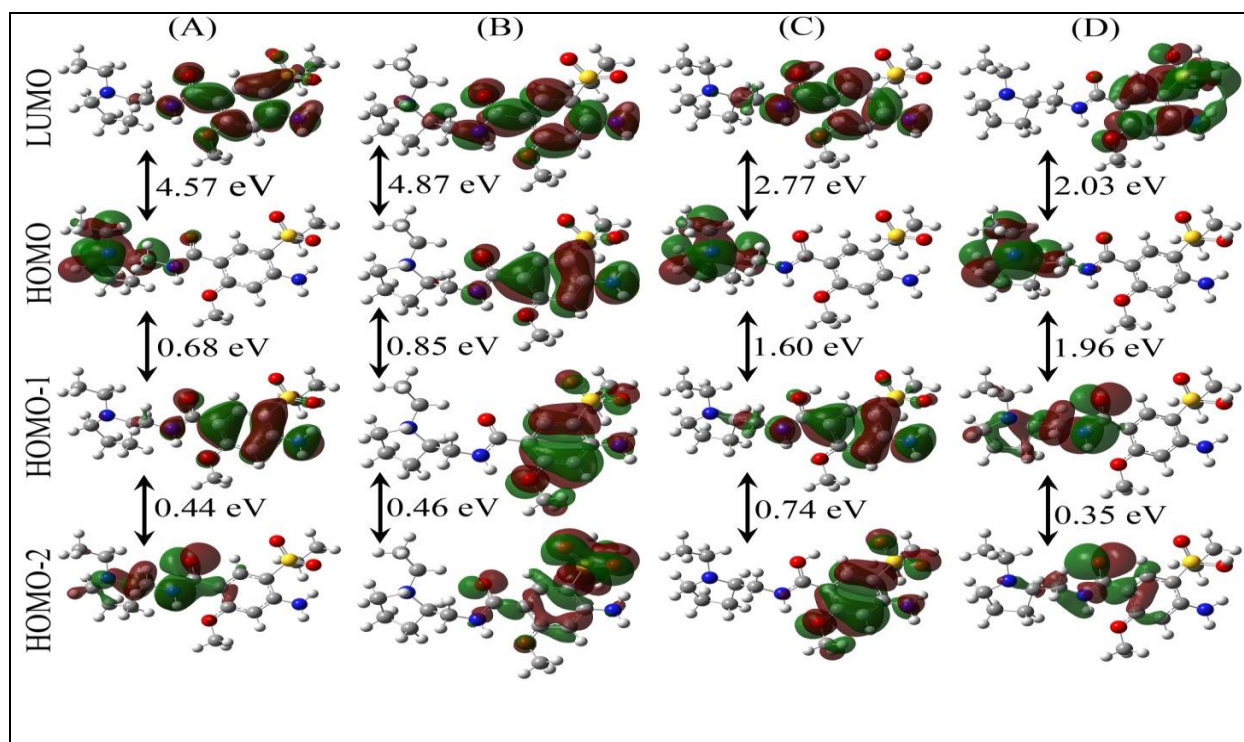


Figure.23 HOMO and LUMO of Amisulpride and its protonated forms

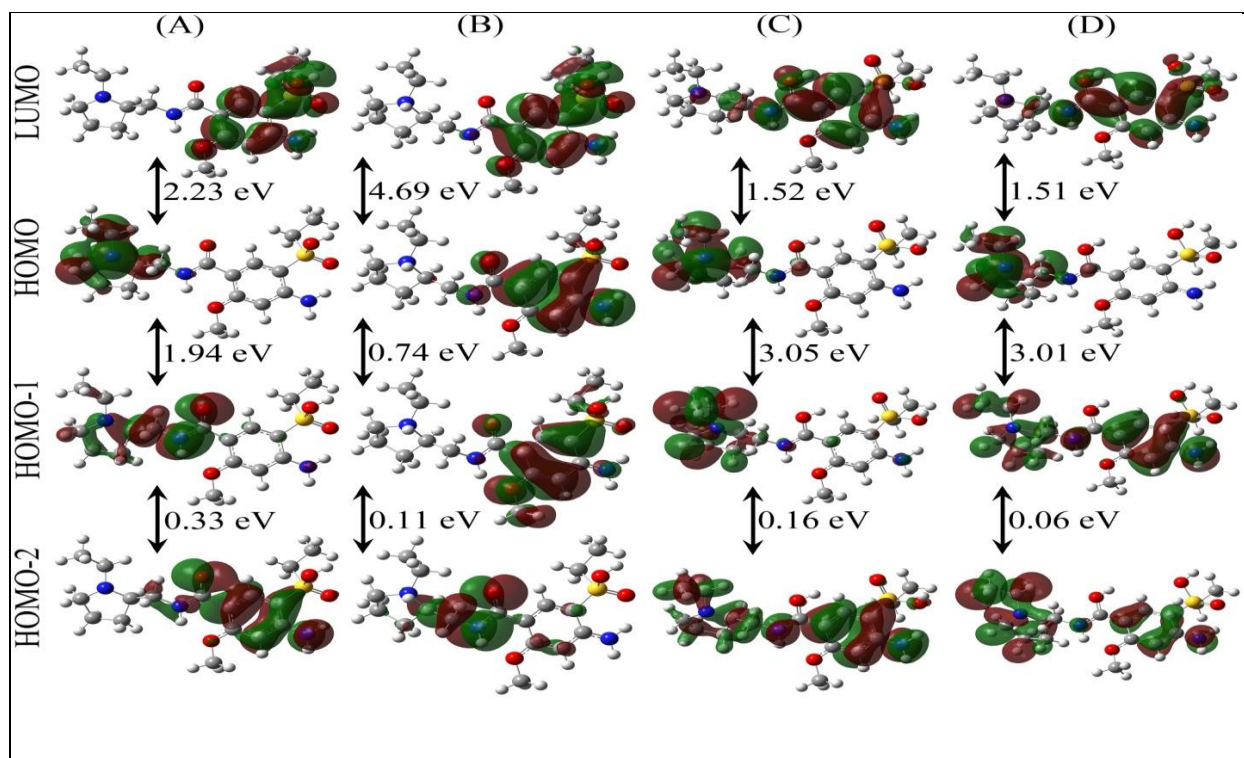


Figure.24 HOMO and LUMO of Amisulpride and its protonated forms

Based on these values, it might be observed that Amisulpride molecule has various HOMO and LUMO

distributions in the neutral form as well as in the protonated form other than (Amisulpride) N_3H^+ and

(Amisulpride)N3O2H2H2+2. The HOMO densities of the Amisulpride, (Amisulpride)O4H +, (Amisulpride)O2H +, (Amisulpride)O1H +, (Amisulpride)O1O4H2+2, (Amisulpride)O2O4H2+2 are centred about the pyrrolidine ring, the ethyl group that is linked to pyrrolidine through its nitrogen atom, and an insignificant portion of the (O4=C9=N. In the case of the LUMO distributions with respect to the structures constituted, it is possible to observe that LUMO is mostly superimposed on the benzene ring and partially on the sulfonyl group and in other cases on the (O4=C9=N2) region. In regard to the (Amisulpride)N 3H + and (Amisulpride)N 3O 2H 2 structure as two exceptions, despite the occupancy of the same side, the HOMO huge is concentrated in the benzene ring and the LUMO huge in the (O4=C 9=N 2) region and sulfoxyl group respectively. Therefore, the main examples of electrons from an inhibitor molecule that can be saved as a coordinate bond is by an copper atom with all electrons in their d orbital number of the atom. Also, the molecule of the inhibitor can take off electrons of an atom of Cu that has its antibonding orbital to develop back-donating bond primarily in its protonated form. Such processes of donation and back-donation enhance the adsorption of various folds of the Amisulpride molecules to the copper surface. Density functional theory (DFT) has been determined to be effective in giving information on the chemical reactivity and selectivity, in both global parameters like electronegativity (χ), hardness (η) and softness (S), and local parameters, including the Fukui function.

($f(r)$) and local softness ($s(r)$) Thereby, the first derivative of the E operator of an N at ($v(r)$) on system with total electronic energy (E) and an external potential ($v(r)$) called chemical potential (μ) negative of electronegativity.

$$\chi = -\left(\frac{\partial E}{\partial N}\right)_{v(r)} = -\mu \quad (7)$$

Hardness (η) has been defined within the DFT as the second derivative of the E with respect to N at property which measures both the stability and reactivity of a molecule.

$$\chi = \left(\frac{\partial^2 E}{\partial N^2}\right) = \left(\frac{\partial \mu}{\partial N}\right)_v \quad (8)$$

E is the electronic energy, N is the number of electrons and v is external potential by the Nuclei and μ is chemical potential. This parameter can be estimated with respect to a HOMO and LUMO molecular orbital energy in accordance with the valence state parabola prediction and with the model of the parabola of most bonding and anti-bonding states.

The quantum chemical perspective also defines the proportion of electronic charge (0 N)

transferred to the metal by the inhibitor as another critical factor. On contact between a bulk metal and an organic corrosion inhibitor, the electron flow will be directed to the atom with lower χ value to the one with higher χ value till the equality of the chemical potentials is achieved. Then, N- came out, or the fraction of charge transferred, Delta N, can be determined:

$$\Delta N = \frac{\chi_M - \chi_I}{2(\eta_M + \eta_I)} \quad (9)$$

where the subscripts M and I represent the metal and inhibitor, respectively. According to Pearson, operational and approximate definitions of the electronic chemical potential (μ) and the absolute hardness (η) of a chemical system are given by:

$$-\mu = \frac{(I+A)}{2} = \chi \quad (10)$$

$$\eta = \frac{(I-A)}{2} \quad (11)$$

where I is the ionization potential and A is the electron affinity. Since $(I+A)/2$ is the Mulliken electronegativity for atoms, the value of χ for any system is known as the absolute electronegativity.

According to Koopman's theorem, the frontier orbital energies are given by:

$$I = E_{\text{HOMO}} \quad (12)$$

$$A = -E_{\text{LUMO}} \quad (13)$$

Recently, (Parr *et al.* 1999; Gomez *et al.* 2006; Robles, *et al.* 2018; Liu *et al.* 2009) have introduced an electrophilicity index (ω) defined as;

$$\omega = \frac{\mu^2}{2\eta} \quad (14)$$

This was proposed as a measure of the electrophilic power of a molecule and global softness is given by;

$$S = 12\eta \quad (15)$$

The values of the calculated theoretical parameters presented in Table 9 will give the information about reactive behavior of Amisulpride molecule and its protonated form. EHOMO is commonly related to the capacity of a given molecule to donate electrons; large values of EHOMO will tend to reflect the ability of the molecule to donate electrons to other suitable acceptor molecules of lower energy molecular orbitals. Instead, ELM is a measure that demonstrates the capacity of the molecule to accommodate electrons. The Hayat of the inhibitor on the metal surface resistance expands with

growing both the HOMO as well as the LUMO energies. Accordingly, there is an increasing likelihood that the molecule would accept electrons as the higher the value of ELUMO. In addition, the difference between the HOMO and LUMO of the molecule is a significant value that defines the reactivity of the inhibitor molecule of the metallic surface adsorption. The smaller the DE (particularly in the case of the cationic species), the more reactive will be the molecule resulting in the growth of the inhibition efficiency of the molecule. To be effective in overlapping a small difference in energy between the orbitals is usually required and the total energy difference between the orbitals (HOMO and HOMO-1) is 0.68 eV in the neutral form of amisulpride, which means that the orbitals are involved in the metal-ligand interaction. Moreover, the energy gap between the HOMO-2 and HOMO-1 of the Amisulpride is so small to confirm that HOMO-1 and HOMO-2 were involved in the interaction between the ligand and the metal (Figure 23). It is also observed that the protonated forms of Amisulpride have the same finding, indicating that chemical reactivity of the parameters of the orbitals of HOMO, HOMO-1 and HOMO- 2 are significant over iron surface. Table 9 clearly demonstrates that in the neutral form, the E HOMO value of Amisulpride molecule is highest and in the protonated form, the E HOMO value is lesser. This implies that the ability of amisulpride to donate electrons is less in the protonated ones. Table 9 has evidenced that protonated form of Amisulpride has the lowest ELUMO, thus the most likely form of interaction of copper on the Amisulpride molecule is its protonated form. As Table 9 indicates, the quantum chemical parameters further indicate that Amisulpride in its protonated form is the least reactive (DE value of 1.511 eV) showing that (Amisulpride) O2O4H2+2 is the most reactive species that can be readily adsorbed on the metal surface leading to greater protection. This is in accordance with the experimental findings that Amisulpride molecule may be superior in its ability to inhibit copper surface in the protonated form that is, through electrostatic interaction between the cationic form of Amisulpride and the free d-orbital of the copper (physisorption). In addition, adsorption of Amisulpride on the steel surface with the neutral form also contributes to the entire process of inhibiting.

Molar volume also provides data about the contacting interface of the inhibitor and the metal. This phenomenon is normally proportional to the percentage of the surface occupied by the adsorbed inhibitor. Molar volume of amisulpride and their structures of protonation follows the pattern Amisulpride > (Amisulpride)O2O4H2+2 > (Amisulpride)O2H+ > (Amisulpride)O1O4H2+2 > (Amisulpride)O1H+ > (Amisulpride)N3H+ > (Amisulpride)O4H+ > (Amisulpride)N3O2H2+2; Table 8.9 also tab Amisulpride molecule exhibits the largest molecular volume in these structures which gives the greatest

coverage area between the molecular inhibitor and surface and the best inhibition efficiency.

The value of DN which is the fraction of copper atoms transferred between the inhibitor and the isotope iron molecule is determined using a χ_M value of 4.8 eV mol⁻¹ and η_M value of 0 eV mol⁻¹ of copper atom. In other reports, the effect of inhibition was observed by values of DN which was a result of electrons donation. In concurrence with the study conducted by Lukovits *et al.* (2001), when DN < 3.6, the higher is the electron-donating ability at the metal surface, the greater the inhibition activity. Moreover, the calculations indicate that in the case of Amisulpride molecule except neutral form, the single protonated forms of Amisulpride molecule have positive DN value that changes to negative value on a second protonation. Therefore, in the current research AMDVs of single protonated Amisulpride is the donor of electrons and the copper surface is the acceptor of electrons and vice versa in the double protonated structures. Electrophilicity index, ω , which is a statistic of the electrophilic strength of a molecule is determined with both neutral and protonated Amisulpride as the form. It is reported that the more the value of ω , the more the similarity the capacity of the molecule to absorb electrons. This research has protonated form of Amisulpride, (Amisulpride) O1O4H2+2, with the highest value of ω and hence the higher ability to receive electrons of the metal. This process enhances the capacity of adsorption of Amisulpride on the copper metal surface. In any corroding system, it is notable that there is inhibitor which is a Lewis base and then there is the metal which is a Lewis acid. One also should take account of the situation that is associated with a molecule that will be receiving some amount of charge in some center and will back-donate some amount of charge in the same center or otherwise. To represent the change of energy following these two reactions, the second order simple charge transfer expression is assumed as a two-parameter equation wherein the two processes, donation and back-donation processes are separated by using the value of the chemical potential of each process, whereas the hardness is set to the value of $(\eta = (\mu^+ - \mu^-))$ in both cases. Therefore, a given charge is added to the molecule, ΔN^+ ;

$$\Delta E^+ = \mu^+ \Delta N^+ + 12\eta(\Delta N^+)^2 \quad (16)$$

While when the molecule back-donates a certain amount of charge, then ΔN^-

$$\Delta E^- = \mu^- \Delta N^- + 12\eta(\Delta N^-)^2 \quad (17)$$

If the total energy change is approximated by the sum of the contributions in Equations 16 and 17, and assuming that the amount of charge back-donated is equal to the amount of charge received, then;

$$\Delta N^- = -\Delta N^+$$

$$\Delta E_T = \Delta E^+ + \Delta E^- = (\mu^+ - \mu^-) \Delta N^+ + \eta (\Delta N^+)^2 \quad (18)$$

The most favorable situation corresponds to the case when the total energy change becomes a minimum with respect to ΔN^+ , which implies that and that;

$$\Delta N^+ = -(\mu^+ - \mu^-) / 2\eta$$

$$\Delta E_T = (\mu^+ - \mu^-)^2 / 4\eta = -\eta^4 \quad (19)$$

Calculations in Table 9 show that electron density (in both normal and protonated) and DET (in both

normal and protonated) of Amisulpride molecule are both/e (larger), and DET (larger) in the neutral and protonated state respectively. This finding means that the charge transfer to Amisulpride molecule and back-donation by the molecule are energetically favorable. It has been reported that similar observation may also be reported. It should however be mentioned that the value of DET obtained does not imply that a back-donation process will take place; it merely confirms the fact that when the same two processes (charge transfer to the molecule and back-donation by the molecule) happen, the energy change is directly proportional to molecule hardness.

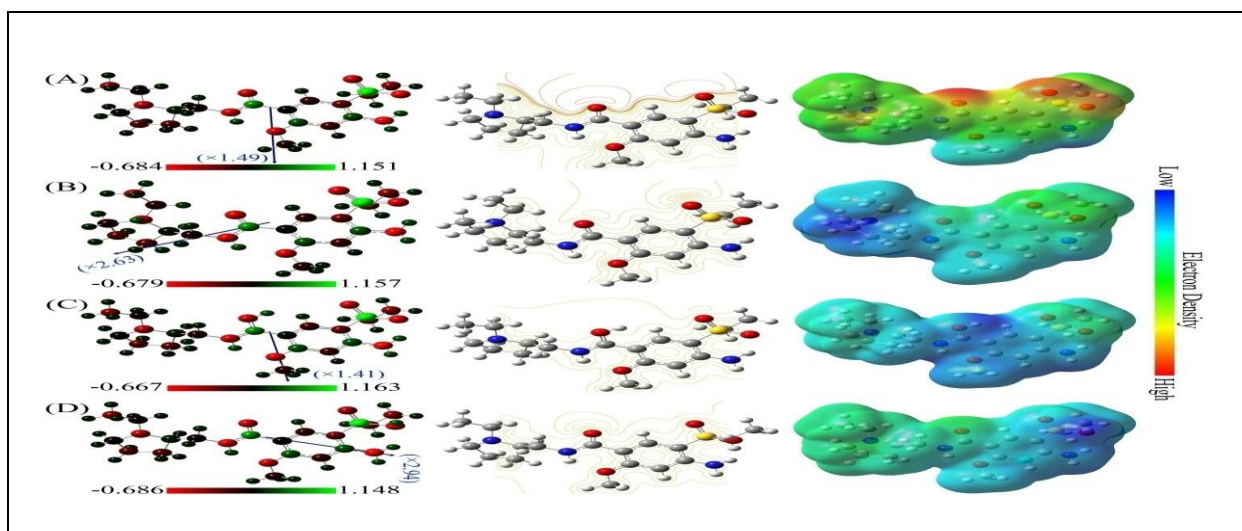


Figure.25 Electrostatic properties of (A) Amisulpride, (B) (Amisulpride)N3H⁺, (C) (Amisulpride)O4H⁺, and (D) (Amisulpride)O1H⁺: Mulliken charge population and the side view of the dipoles are displayed on the left while the middle and right panels show 2D contour map and 3D MESP surface of these structures, respectively.

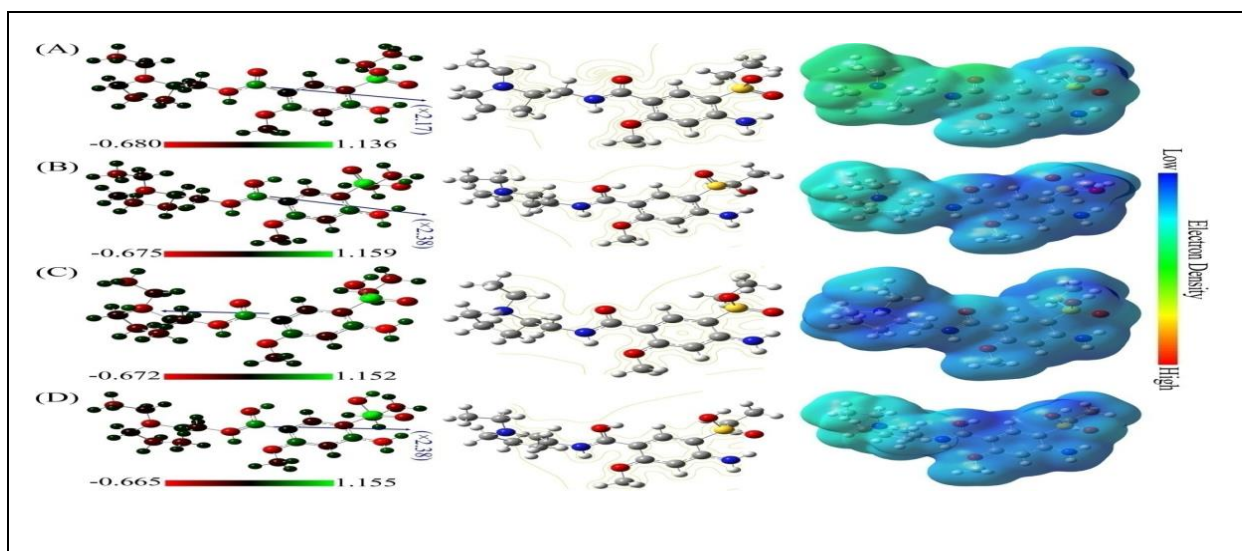


Figure.26 Electrostatic properties of (A) (Amisulpride)O₂H⁺, (B) (Amisulpride)O₁₀H₂²⁺, (C) (Amisulpride)N₃O₂H₂²⁺, and (D) (Amisulpride)O₂O₄H₂²⁺: Mulliken charge population and the side view of the dipoles are displayed on the left while the middle and right panels show 2D contour map and 3D MESP surface of these structures, respectively.

It may be pointed out in this work that besides the (Amisulpride)O₄H⁺, (Amisulpride)N₃O₂H₂²⁺, other protonated structures have higher values of dipole moment compared to the unprotonated form of Amisulpride. The direction of the dipole may be explained using electrostatic potential (middle and right panels of Figure 25 and 26.), which identifies the electrons density rich regions that are concentrated around the said atoms.

4. CONCLUSIONS

This research paper has made it evident that the expired drug Amisulpride is an effective copper corrosion inhibitor in 0.5 M H₂SO₄ and 1.0 M HNO₃ solutions. Weight-loss, EIS and polarization analyses were invariably found to show that the inhibition efficiency peaked at the inhibitor concentration and went to maximum with the change in concentration of 1.0 mM where a significant decrease in the corrosion rate was realised. The enhancement in the resistance of charge transfer and the reverse lowered capacitance of the two layers indicates the presence of a protective layer of adsorption on the surface of copper. A polarization study revealed that Amisulpride was a mixed-type inhibitor with more pronounced inhibition of the cathodic process that indicates its capacity to suppress the reaction of hydrogen evolution.

Adsorption experiments proved that Amisulpride obeys the Langmuir adsorption isotherm, which implies that the process occurs in the monolayer form but the negative, negative ΔG° values prove that adsorption is spontaneous and predominantly physisorption with a small portion of chemisorption. Due to the high temperatures, the coverage of the surface was increasing and the decrease in activation energy of the reaction in the presence of the inhibitor continues to

substantiate a high degree of interaction between Amisulpride and the metal surface. Quantum chemical calculations were also helpful in understanding the molecular mechanism wherein both neutral and protonated forms of Amisulpride have favourable electronic properties which allow them to make adsorption via electron donation and back-donation mechanisms.

In general, the results exhibit that the expired Amisulpride may be successfully utilized as an environmentally friendly, sustainable, and inexpensive copper corrosion inhibitor in the aggressive media with low acidity. This combination of experimental and theoretical modelling makes it more probable that it can be applied in industrial systems where corrosion control is necessary.

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CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

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